PRODUCT REQUIREMENTS DOCUMENT

PRD v1.0

BrandSphere AI

AI-Powered Automated Branding Assistant for Businesses

Scenario 1 — CRS Artificial Intelligence Capstone Project 2025–26

Course: Artificial Intelligence | Type: Capstone (Group) | Marks: 100

Stack: Python (Google Colab) • Gemini API • Streamlit Cloud • Tableau / Plotly • OpenCV • TensorFlow/Keras

Prepared By: Student Group — IADAI205 -2405728-Aashi-Dimple-Soni -2405426-Hemer-Manish-Pandya -2405744-Zivantika-Amit-Singh-	**Version:** 1.0 — Final Submission
Document Status: **Complete**	**Academic Year:** 2025 – 2026

TABLE OF CONTENTS

1. **Executive Summary**3
2. Project Overview & Problem Statement4
3. Intended Learning Outcomes (ILOs)5
4. System Architecture & Technology Stack6
5. Datasets & Data Pipeline7
6. Core Features & Module Specifications9
7. AI Model Design & Training Plan13
8. UI/UX Design & Streamlit Interface15
9. Visualization & Analytics Dashboard16
10. Feedback Intelligence & Model Refinement17
11. 10-Week Project Timeline18
12. GitHub Repository Structure20
13. Evaluation Rubric & Evidence Checklist21
14. Risk Register & Mitigation23
15. Ethical AI & Responsible Design24
16. Conclusion25

1. EXECUTIVE SUMMARY

BrandSphere AI is an end-to-end intelligent branding platform developed as the capstone project for the CRS Artificial Intelligence course (2025–26). The platform enables small and medium-sized businesses (SMBs) to build a complete brand identity autonomously using Artificial Intelligence, Computer Vision, and Natural Language Processing.

The system integrates five tightly coupled AI modules: (1) an AI Logo & Design Studio that auto-generates brand visuals; (2) a Creative Content & GenAI Hub that produces multilingual taglines and animated assets via the Gemini API; (3) a Smart Social & Brand Campaign Studio that forecasts engagement KPIs and generates social-media-ready content; (4) a Brand Aesthetics Engine that enforces visual and tonal consistency; and (5) a Feedback Intelligence Loop that enables continuous model improvement.

Platform Name	BrandSphere AI — Automated Branding Assistant
Scenario	Scenario 1: AI-Powered Automated Branding Assistant for Businesses
Problem	SMBs lack affordable, expert branding tools — leading to inconsistent, low-impact brand identities
Solution	AI-driven platform that auto-generates logos, slogans, color palettes, campaigns, and analytics
Tech Stack	Python, Gemini API, TensorFlow/Keras, OpenCV, scikit-learn, Streamlit Cloud, Plotly/Tableau
Duration	10 Weeks | Culminating in a 30-minute WACP Panel Presentation + Live Demo

2. PROJECT OVERVIEW & PROBLEM STATEMENT

2.1 Background

Branding is the single most critical differentiator for any business. Yet over 70% of SMBs globally operate without a coherent brand identity due to the prohibitive cost of professional designers and marketing agencies. Generic templates produce cookie-cutter outputs that fail to reflect a company's personality, audience, or industry.

2.2 Problem Statement

Small and medium-sized businesses require a fast, affordable, and intelligent solution that can:

- Automatically generate custom logos, fonts, and color palettes aligned with brand personality.
- Produce AI-crafted taglines, slogans, and brand narratives in multiple languages.
- Generate ready-to-publish social media campaigns with engagement performance predictions.
- Ensure visual and tonal consistency across all marketing materials.
- Learn from user feedback to continuously improve outputs over time.

2.3 Proposed Solution

BrandSphere AI directly addresses all five pain points through an integrated AI pipeline:

1. Logo & Design Studio — CV + Generative AI for visual brand assets.
2. Creative Content Hub — Gemini API for multilingual, tone-adaptive NLP outputs.
3. Campaign Studio — ML-based CTR/ROI forecasting + social content generation.
4. Brand Aesthetics Engine — Semantic similarity + color psychology for consistency.
5. Feedback Intelligence — Reinforcement learning loop for iterative improvement.

2.4 Target Users

User Segment	Primary Need	Key Benefit
SMB Founders	Brand identity from scratch	End-to-end kit in minutes
Marketing Teams	Campaign content at scale	AI-generated multilingual campaigns
Freelance Designers	Brand consistency tools	Color + font + slogan alignment
Startup Accelerators	Rapid prototype branding	Batch brand kit generation

3. INTENDED LEARNING OUTCOMES (ILOs)

This project is evaluated against three primary Intended Learning Outcomes (ILOs) as defined by the CRS AI Capstone brief:

#	ILO Statement	How BrandSphere AI Demonstrates It
ILO 1	Demonstrate the ability to design and develop AI-powered applications that deliver personalized, real-time, and intelligent solutions in creative and data-driven industries.	Personalized logo, slogan, and campaign generation based on user inputs using trained ML + Gemini models.
ILO 2	Integrate AI models into interactive and visually engaging applications that enhance user experience through automation and analytics.	Streamlit-based interactive UI with real-time AI output previews, animated GIFs, and Plotly dashboards.
ILO 3	Apply proficiency in deploying AI-based applications that connect backend data processing with intuitive, user-centric front-end designs.	Full deployment on Streamlit Cloud with Python Colab backend, GitHub CI integration, and downloadable ZIP campaign kits.

4. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

4.1 High-Level Architecture

BrandSphere AI follows a three-tier architecture: (1) Data & Model Layer, (2) AI Processing Engine, and (3) Frontend Interface.

ARCHITECTURE FLOW DIAGRAM

[USER INPUT via Streamlit UI]

↓ Company Name / Industry / Brand Personality / Target Audience

[AI PROCESSING ENGINE — Google Colab / Python Backend]

```
┌──────────────────────────────────────────────────────────┐
│ Module 1: Logo Studio (CNN + OpenCV + TensorFlow)        │
│ Module 2: GenAI Hub (Gemini API + NLTK)                  │
│ Module 3: Campaign Studio (scikit-learn + Gemini API)    │
│ Module 4: Brand Aesthetics Engine (Sentence Transformers) │
│ Module 5: Feedback Loop (RL / Fine-tuning + Google Drive DB) │
└──────────────────────────────────────────────────────────┘
```

[OUTPUT] → Streamlit UI Preview + Downloadable ZIP Campaign Kit

[STORAGE] → Google Drive / Colab + GitHub Repository

4.2 Technology Stack

Layer	Technology	Purpose
AI / ML Backend	TensorFlow 2.x / Keras	CNN for logo classification & font recognition
AI / ML Backend	scikit-learn	Campaign KPI prediction (CTR, ROI, Engagement)
AI / ML Backend	OpenCV	Image processing, color palette extraction
Generative AI	Google Gemini API (gemini-1.5-flash)	Tagline, slogan & multilingual content generation
NLP	NLTK / Sentence Transformers	Text preprocessing, semantic similarity scoring
Data Processing	pandas / NumPy	Dataset loading, cleaning, feature engineering

Visualization	Plotly / Tableau Public	Interactive analytics dashboards
Frontend / UI	Streamlit Cloud	Deployed interactive web application
Animation	Pillow / imageio	Animated GIF generation for brand assets
Development	Google Colab / Jupyter	Model training and experimentation
Version Control	GitHub	Code repository & collaborative development
Storage	Google Drive	Feedback data storage & model checkpoints

5. DATASETS & DATA PIPELINE

5.1 Required Datasets (Provided by CRS)

Dataset	Description	Used In Module	Key Features
Logo Dataset	Brand logo images across industries with classification labels	Module 1: Logo & Design Studio	Industry label, style tags, dominant color, resolution
Font Dataset	Font image samples with metadata (serif/sans-serif, style, weight)	Module 1: Logo & Design Studio	Font category, visual weight, use-case tag
Slogan Dataset	Thousands of brand slogans with industry, tone, and audience tags	Module 2: GenAI Hub	Tone, industry, target audience, language
Startups Dataset	Startup profiles with brand attributes, industry, and persona data	Modules 2, 4	Industry vertical, brand persona, founding year, stage
Marketing Campaign Dataset	Historical campaigns with CTR, ROI, and Engagement Score labels	Module 3: Campaign Studio	Platform, region, objective, CTR, ROI, Engagement Score

5.2 Data Pipeline

All datasets pass through a standardized preprocessing pipeline before being used for model training:

Step	Stage	Activities
1	Data Ingestion	Load raw CSV, image folders, and JSON files into Google Colab using pandas and os.walk()
2	Exploratory Data Analysis (EDA)	Visualize class distributions, check for nulls, analyze feature correlations, inspect image quality
3	Data Cleaning	Remove duplicates, handle missing values (imputation/drop), standardize column names, fix encoding issues
4	Feature Engineering	Extract color histograms, image embeddings (CNN), text TF-IDF/embeddings, one-hot encode categoricals
5	Train/Val/Test Split	80/10/10 stratified split ensuring class balance across all supervised models

6	Data Augmentation (Image)	Apply rotation, flip, brightness adjustment, and zoom using Keras ImageDataGenerator
7	Normalization / Scaling	MinMaxScaler for tabular features, pixel normalization [0,1] for image data
8	Cleaned Dataset Export	Save processed datasets to Google Drive and /datasets/ GitHub folder

5.3 EDA Deliverables

The EDA Jupyter notebook (notebooks/01_EDA.ipynb) will include:

- Dataset shape, dtypes, null count, and unique value analysis for all 5 datasets.
- Distribution plots for campaign KPIs (CTR, ROI, Engagement Score) using seaborn/Plotly.
- Word clouds of most frequent slogans and brand terms from the Slogan and Startups datasets.
- Color frequency heatmaps extracted from the Logo Dataset using KMeans clustering.
- Confusion matrix and class-balance charts for font and logo classification tasks.
- Correlation matrix for marketing campaign tabular features to guide feature selection.

6. CORE FEATURES & MODULE SPECIFICATIONS

Module 1: AI Logo & Design Studio

Objective: Automatically generate logos, recommend suitable fonts, and extract color palettes aligned with the company's brand personality and industry.

Sub-Feature	Specification
Logo Style Classifier	CNN (VGG16 transfer learning) trained on Logo Dataset to classify logos by industry and style (minimalist, vibrant, luxury). Output: Top-3 matched style clusters.
Color Palette Extractor	KMeans (k=5) on pixel RGB values using OpenCV. Output: 5-color HEX palette with psychology labels (e.g., trust=blue, energy=orange). Displayed as swatches in Streamlit.
Font Recommender	CNN or KNN model trained on Font Dataset. Recommends Top-3 fonts based on brand personality input. Renders live font preview in Streamlit.
Brand Visual Preview	Unified brand card showing generated logo + color swatches + font sample, rendered as a PNG using Pillow. Saved to campaign ZIP package.

User Inputs: Company name, industry (dropdown), brand personality (minimalist / vibrant / luxury / bold / elegant).

Module 2: Creative Content & GenAI Hub

Objective: Generate brand narratives, taglines, animated visuals, and multilingual content using the Gemini API.

Sub-Feature	Specification
Tagline Generator	Gemini API (gemini-1.5-flash) prompted with company name, target audience, and tone. Returns 5 tagline variants ranked by brand fit score from Slogan Dataset embeddings.
Brand Narrative Writer	Generates a 150-word brand story and 3 creative brand messages. Uses few-shot prompting with Startups Dataset examples to tune tone consistency.
Multilingual Translator	Gemini API translates generated taglines into 5 languages: Hindi, French, Spanish, Arabic, Mandarin. Output displayed as a language card grid in Streamlit.
Animated GIF Creator	Pillow + imageio creates a 5-frame animated GIF: logo fade-in, tagline typewriter effect, color palette sweep. Embedded in Streamlit and saved to campaign ZIP.

User Inputs: Company name, target audience description, preferred tone (formal / bold / youthful / inspirational).

Module 3: Smart Social & Brand Campaign Studio

Objective: Generate professional marketing campaigns with social content, performance predictions, and downloadable ZIP output.

Sub-Feature	Specification
Social Content Generator	Gemini API generates Instagram/Facebook/Twitter post captions (280 chars), hashtag sets (15 tags), and regional strategy summaries based on platform + objective + region inputs.
KPI Predictor	Random Forest Regressor trained on Marketing Campaign Dataset. Predicts CTR (%), ROI (%), and Engagement Score (0–100) for proposed campaigns. RMSE and R^2 reported.
Location Insights	Region-wise campaign performance analysis visualized as Plotly choropleth maps showing predicted engagement by geography.
Campaign ZIP Export	Packages all outputs (logo PNG, color palette JSON, taglines TXT, captions TXT, hashtags TXT, analytics PNG, translations CSV) into a downloadable ZIP using Python zipfile.

User Inputs: Platform (Instagram / Facebook / Twitter/X), target region, campaign objective, product/service description.

Module 4: Brand Aesthetics Engine

Objective: Ensure visual and tonal consistency across all AI-generated brand assets using color psychology and semantic similarity.

- Color Harmony Scorer: Evaluates uploaded/generated palette against color psychology rules (complementary, triadic, analogous). Score displayed 0–100.
- Font-Tone Aligner: Sentence Transformers compute cosine similarity between tagline text embeddings and font metadata descriptions to ensure tonal alignment.
- Brand Consistency Report: Automated PDF summary (ReportLab) showing logo, palette, fonts, slogans, and alignment scores.
- Improvement Recommendations: AI-suggested alternative colors or fonts if consistency score falls below 70%.

Module 5: Feedback Intelligence & Model Refinement

Objective: Collect user ratings and textual feedback for all generated outputs, and use this data to iteratively improve model performance.

- Rating Widget: Streamlit star-rating (1–5) for each generated element: logo, tagline, campaign, palette.
- Feedback Storage: Feedback JSON stored on Google Drive via Google Drive API with timestamp and session ID.
- Sentiment Analysis: TextBlob applied to textual comments to classify feedback as positive/neutral/negative.
- Feedback Analytics Dashboard: Plotly bar charts showing average ratings per module, most-improved assets, and feedback volume over time.
- Model Retraining: Colab notebook (notebooks/05_retraining.ipynb) for fine-tuning models using high-rated samples and down-weighting poor-rated samples via reinforcement learning reward shaping.

7. AI MODEL DESIGN & TRAINING PLAN

Model	Architecture	Dataset	Task	Metric
Logo Classifier	VGG16 Transfer Learning (Keras)	Logo Dataset	Multi-class classification (industry/style)	Accuracy, F1
Font Recognizer	CNN + KNN hybrid	Font Dataset	Font family classification	Top-3 Accuracy
Color Extractor	KMeans (k=5, scikit-learn)	Logo Dataset (pixels)	Dominant color extraction	Inertia, Silhouette
Tagline Generator	Gemini API (few-shot prompted)	Slogan Dataset	Creative text generation	BLEU, User Rating
Campaign KPI Model	Random Forest Regressor	Marketing Campaign Dataset	CTR / ROI / Engagement regression	RMSE, R^2
Brand Consistency	Sentence Transformers (SBERT)	Startups + Slogan Datasets	Semantic similarity scoring	Cosine Similarity
Feedback Sentiment	TextBlob + Gemini summarizer	Collected feedback data	Sentiment classification	Accuracy, VADER F1

7.1 Model Training Protocol

- All models trained in Google Colab using GPU runtime (T4 or A100).
- Experiment tracking: log loss curves, accuracy plots, and confusion matrices saved to /notebooks/ folder.
- Model checkpoints saved to Google Drive every 5 epochs to prevent data loss.
- Final trained model weights saved as .h5 (Keras) or .pkl (scikit-learn) and committed to GitHub.
- Hyperparameter tuning: GridSearchCV for scikit-learn models; manual LR scheduling for CNNs.

7.2 Gemini API Integration Details

The platform uses Google Gemini API (gemini-1.5-flash) for the following generative tasks:

Use Case	Prompt Strategy	Output Format
Tagline generation	Few-shot: 3 examples from Slogan Dataset + brand inputs	5 tagline variants as JSON array
Brand narrative	Role prompting: 'Act as senior copywriter for {industry}'	150-word brand story + 3 messages

Multilingual translation	Structured prompt: translate to [language list]	Language-keyed JSON dictionary
Social media captions	Platform-specific formatting rules in system prompt	Platform-specific text + hashtags
Feedback summarization	Summarize N feedback items into actionable insights	3-point improvement summary

8. UI/UX DESIGN & STREAMLIT INTERFACE

8.1 Design Principles

- Progressive Disclosure: Show inputs first; reveal outputs inline without page reload.
- Modular Navigation: Streamlit st.tabs() for each module — keeps interface clean and focused.
- Responsive Layout: st.columns() for side-by-side comparison of generated assets.
- Download Centralization: Single 'Download Campaign Kit' button at end of workflow (ZIP).
- Visual Consistency: Custom CSS injected via st.markdown() for brand-aligned UI colors.

8.2 Streamlit App Structure

Tab	Module	Key Streamlit Components
🏠 Home	Onboarding + Brand Input Form	st.text_input, st.selectbox, st.multiselect — collects company name, industry, personality
🎨 Design	Logo & Design Studio	st.image, st.color_picker, st.download_button — logo preview + palette swatches
✍️ Content	Creative Content Hub	st.write, st.expander, st.metric — taglines, brand story, animated GIF embed
📣 Campaign	Campaign Studio	st.plotly_chart, st.dataframe, st.download_button — KPI predictions + campaign content
🔍 Aesthetics	Brand Aesthetics Engine	st.progress, st.alert — consistency score + recommendations
⭐ Feedback	Feedback Intelligence	st.slider, st.text_area, st.form — ratings + textual feedback collection
📊 Analytics	Dashboard	st.plotly_chart, Tableau embed — feedback trends + campaign analytics

8.3 Wireframe Descriptions

Full wireframes will be designed in Figma and stored in /ui_ux/wireframes/. Key screens:

- Home Screen: Hero banner with BrandSphere logo, 5-field input form, CTA button.
- Design Output Screen: Split layout — left: generated logo + palette; right: font recommendations.
- Campaign Screen: 3-column layout — content | KPI gauges | regional map.
- Feedback Screen: Module selector dropdown + star rating widget + text comment box.

9. VISUALIZATION & ANALYTICS DASHBOARD

BrandSphere AI includes dedicated Plotly (embedded in Streamlit) and Tableau Public dashboards:

Chart / Visual	Tool	Description
Campaign KPI Bar Chart	Plotly	Predicted CTR, ROI, and Engagement Score displayed as grouped bar chart with benchmark overlay
Color Palette Pie Chart	Plotly	Proportion of each extracted dominant color in the brand palette
Regional Engagement Map	Plotly Choropleth	World map color-coded by predicted campaign engagement by target region
Feedback Rating Trends	Plotly Line Chart	Average rating per module tracked over time (session history)
Sentiment Distribution	Plotly Donut Chart	Positive / Neutral / Negative distribution of user feedback comments
Slogan Similarity Heatmap	Plotly Heatmap	Cosine similarity matrix of generated taglines vs. Slogan Dataset top entries
Tableau Campaign Dashboard	Tableau Public	Embedded Tableau dashboard: campaign ROI by industry, CTR trends, and audience engagement breakdown
Logo Style Cluster Visual	Plotly Scatter (UMAP)	UMAP 2D visualization of Logo Dataset clusters colored by style category

10. FEEDBACK INTELLIGENCE & MODEL REFINEMENT

10.1 Feedback Collection Architecture

Every AI-generated output in BrandSphere AI includes an embedded feedback widget. Feedback data is structured as a JSON record:

Field	Type	Example
session_id	UUID string	a3f7-4b2c-9d1e-...
timestamp	ISO 8601 datetime	2025-11-01T14:32:00Z
module	Enum string	logo_studio | content_hub | campaign_studio
output_id	Hash string	sha256 of generated output
star_rating	Integer 1–5	4
comment	Free text string	The logo style is great but fonts feel generic
sentiment	Enum string	positive | neutral | negative
preferred_alternative	String (optional)	More vibrant color palette

10.2 Model Refinement Loop

6. Feedback records (min. 50 samples) aggregated weekly from Google Drive.
7. High-rated samples (star_rating >= 4) extracted as positive training examples.
8. Low-rated samples (star_rating <= 2) flagged for reweighting or exclusion.
9. Retraining notebook (05_retraining.ipynb) run in Colab to fine-tune models.
10. Updated model weights committed to GitHub and deployed to Streamlit Cloud.
11. A/B comparison: feedback analytics before vs. after retraining displayed in dashboard.

11. 10-WEEK PROJECT TIMELINE

Week	Phase	Activities & Deliverables	Evidence
1	Project Scoping	Read brief, select Scenario 1, form group, create GitHub repo, draft PRD v0.1 outline, register Gemini API keys	PRD v0.1 draft + GitHub repo created
2	Data Acquisition & EDA	Download all 5 datasets, run EDA notebook, generate distribution plots, word clouds, identify data quality issues	EDA notebook (01_EDA.ipynb) committed
3	Data Preprocessing	Clean datasets, engineer features, build augmentation pipeline, export cleaned datasets to /datasets/ folder	Cleaned datasets + preprocessing notebook
4	Module 1: Logo & Design	Train VGG16 logo classifier, build KMeans color extractor, train font KNN model, test in Colab	Logo classifier + color extractor notebooks
5	Module 2: GenAI Hub	Integrate Gemini API for tagline/narrative generation, build multilingual translator, create animated GIF pipeline	GenAI Hub notebook + sample outputs
6	Module 3: Campaign Studio	Train Random Forest KPI predictor, build Gemini social content generator, create regional analytics maps	Campaign model notebook + Plotly visualizations
7	Modules 4 & 5	Build Brand Aesthetics Engine (SBERT), integrate feedback widget, set up Google Drive storage, build feedback dashboard	Aesthetics + feedback notebooks + dashboard
8	Streamlit Frontend	Build complete Streamlit app with all 7 tabs, integrate all modules, implement ZIP download feature, apply custom CSS	app.py + Streamlit local demo video
9	Tableau Dashboard + Testing	Create Tableau Public dashboard, perform end-to-end testing, fix bugs, optimize performance, user acceptance testing	Tableau dashboard URL + test report
10	Deployment & PRD Final	Deploy to Streamlit Cloud, finalize PRD PDF, complete GitHub README, rehearse 30-min panel presentation	Live Streamlit URL + PRD PDF + README

12. GITHUB REPOSITORY STRUCTURE

12.1 Repository Naming Convention

IADAI205-(Student1_ID)-Student1Name-(Student2_ID)-Student2Name-(Student3_ID)-Student3Name-(Student4_ID)-Student4Name

12.2 Folder Structure

Path	Contents
/datasets/	original/ — raw downloaded datasets; cleaned/ — preprocessed CSVs/images ready for model training
/notebooks/	01_EDA.ipynb, 02_preprocessing.ipynb, 03_logo_model.ipynb, 04_campaign_model.ipynb, 05_retraining.ipynb
/ui_ux/wireframes/	Figma exported PNGs/PDFs of all screen wireframes and Streamlit prototype screenshots
/ui_ux/assets/	BrandSphere logo, icons, background images used in Streamlit UI
/config/	requirements.txt (all Python dependencies), docs/ (API docs, model cards)
/deployment/	streamlit_deployment.yml, .streamlit/config.toml, Procfile (if needed)
app.py	Main Streamlit application file — all tabs and module integrations
utils/	Helper Python modules: gemini_helper.py, logo_model.py, campaign_model.py, feedback.py
README.md	Full project documentation per checklist (see Section 13)
PRD_BrandSphereAI.pdf	This document — Final Product Requirements Document

13. EVALUATION RUBRIC & EVIDENCE CHECKLIST

13.1 Marks Allocation (Total: 100 Marks)

#	Evaluation Criterion	Marks	Evidence Required
1	PRD Quality & Completeness	15	This document — all sections complete, professional formatting, clear architecture diagrams
2	Dataset Handling & EDA	10	EDA notebook with 6+ visualizations, data quality analysis, preprocessing pipeline documented
3	AI Model Development (Modules 1–3)	25	Trained models committed to GitHub, training notebooks, metrics (accuracy, RMSE, F1) reported
4	GenAI Integration (Gemini API)	10	Working tagline generator, multilingual output, animated GIF creation demonstrated in app
5	Streamlit Deployment & UI/UX	15	Live Streamlit Cloud URL, all 7 tabs functional, Figma wireframes in repo
6	Visualization Dashboards	10	Plotly charts in app + Tableau Public dashboard URL shared in README
7	Feedback Intelligence System	10	Rating widgets working, feedback stored to Drive, analytics dashboard showing trends
8	GitHub Repository Quality	5	Clean folder structure, detailed README, requirements.txt, no API keys in code
TOTAL		**100**	

13.2 Final Submission Checklist

✓	Deliverable	Location
☐	PRD Document (this file) as PDF	GitHub root + Submission portal
☐	EDA Jupyter Notebook	/notebooks/01_EDA.ipynb
☐	Preprocessing Notebook	/notebooks/02_preprocessing.ipynb
☐	Logo & Font Model Notebook	/notebooks/03_logo_model.ipynb
☐	Campaign KPI Model Notebook	/notebooks/04_campaign_model.ipynb

☐	Model Retraining Notebook	/notebooks/05_retraining.ipynb
☐	Original & Cleaned Datasets	/datasets/original/ and /datasets/cleaned/
☐	Streamlit App Code	/app.py + /utils/
☐	Streamlit Cloud Live URL	In README.md
☐	Tableau Dashboard URL	In README.md + /deployment/
☐	Figma Wireframes	/ui_ux/wireframes/
☐	requirements.txt	/config/requirements.txt
☐	README.md (full checklist)	GitHub root
☐	GitHub Repository (naming convention)	Public, accessible to assessors
☐	30-min Presentation (PRD + Live Demo)	Panel session — Week 10

14. RISK REGISTER & MITIGATION

Risk	Likelihood	Impact	Mitigation Strategy
Gemini API rate limits exceeded during demo	Medium	High	Cache API responses in JSON files; use mock outputs for demo if needed
Logo Dataset class imbalance affects CNN accuracy	High	Medium	Apply class weights + data augmentation; use F1 (weighted) instead of accuracy
Streamlit Cloud free tier resource limits	Medium	Medium	Optimize model loading with @st.cache_resource; use lightweight model variants
Google Drive API authentication issues in deployment	Low	High	Use Streamlit Secrets for credentials; test OAuth flow 2 weeks before submission
Team member unavailability near deadline	Low	High	Assign backup roles; maintain shared progress log; weekly check-ins
Poor feedback volume (< 50 samples) for retraining	Medium	Low	Seed feedback dataset with 20 synthetic samples to bootstrap the learning loop
Tableau Public dashboards going offline	Low	Medium	Export Tableau as PNG backup; embed Plotly as primary; Tableau as bonus

15. ETHICAL AI & RESPONSIBLE DESIGN

15.1 Data Privacy & Security

- No personally identifiable information (PII) is collected by BrandSphere AI.
- All Gemini API keys are stored as Streamlit Secrets — never hardcoded in source files.
- Feedback data is anonymized using session UUIDs before storage to Google Drive.
- GitHub repository will not contain any credentials, API keys, or private dataset content.

15.2 AI Fairness & Bias Mitigation

- Logo Dataset will be audited for industry representation bias; underrepresented classes up-sampled.
- Slogan Dataset reviewed for culturally insensitive language before model training.
- Multilingual output quality tested across all 5 languages by native speakers (where possible).
- Campaign KPI model evaluated separately for different industries to detect performance disparity.

15.3 Transparency & Explainability

- All AI-generated outputs are labeled 'AI-Generated' in the Streamlit interface.
- Model cards committed to /config/docs/ describing training data, intended use, and limitations.
- Confidence scores displayed alongside predictions (KPI model, logo classifier).
- Feedback system enables users to flag inappropriate or poor-quality AI outputs.

16. CONCLUSION

BrandSphere AI represents a comprehensive, industry-aligned demonstration of end-to-end AI product development. By combining Computer Vision, Natural Language Processing, Generative AI, Predictive Analytics, and modern deployment practices, this platform delivers tangible value to small and medium-sized businesses while showcasing the full spectrum of skills developed throughout the CRS AI course.

The 10-week development roadmap is structured to ensure progressive, evidence-based delivery — from data exploration and model training through to a fully deployed Streamlit application with Tableau analytics and an intelligent feedback loop. Every component of the platform is directly mapped to the assignment rubric, the three Intended Learning Outcomes, and the GitHub evidence checklist.

BrandSphere AI is not merely a student project — it is a production-ready prototype that demonstrates how AI can democratize professional branding for businesses of all sizes. The team is committed to delivering a polished, ethical, and technically rigorous solution by Week 10.

Document Information	Prepared By
Title: BrandSphere AI — PRD v1.0	Student Group — IADAI205
Scenario: 1 — AI-Powered Automated Branding Assistant	Submitted to: CRS Facilitators + WACP Panel
Course: CRS Artificial Intelligence Capstone 2025–26	Presentation: Week 10 — 30-minute Live Demo
Marks: 100 | Type: Group Capstone	Stack: Python • Gemini API • Streamlit • Tableau

Plan it like a Product Manager. Build it like an AI Engineer.